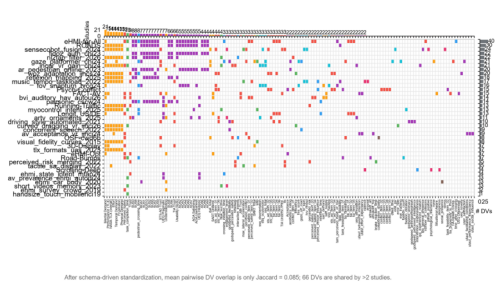
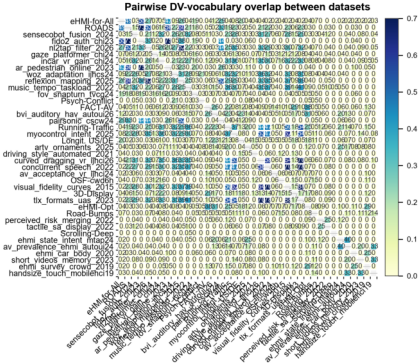
OpenDV-HCI: standardizing dependent variables turns fragmented open datasets into comparable, synthesizable evidence

(a) DV coverage across 39 datasets

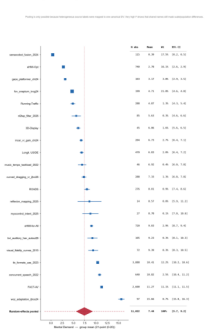


After schema-driven standardization, mean pairwise DV overlap is only Jaccard = 0.085; 66 DVs are shared by >2 studies.

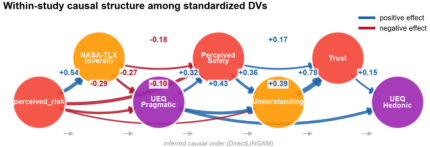
(b) Pairwise overlap (Jaccard)



(c) Meta-analysis: Mental Demand



(d) Cross-study causal structure



DirectLiNGAM on within-study z-scores (complete-case on the largest co-observed DV set; n = 270 obs; all rows from one study (Running-Traffic), i.e. a within-study DAG). Edge weights are structural coefficients.